



BACHELOR IN INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION DECEMBER 2025

Course : EC3103 (Introduction to Information
Systems)

Lecturer: Muhammad Hamizan Johari

Time: 2.00 pm – 5.00 pm (3 hours)

Date : 18 February 2026

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 3 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1.
 - a. Briefly describe **TWO (2)** types of transaction process. (6 MARKS)
 - b. Briefly discuss the advantages of IT careers. (4 MARKS)
 - c. Becoming an agile company is essential in IT because it helps organizations keep pace with fast-moving technology and shifting customer needs. By emphasizing iterative delivery, regular customer feedback, and cross-functional teamwork, agile ways of working enable teams to ship high-quality, innovative solutions faster. This flexibility helps businesses remain competitive, seize market opportunities quickly, and consistently deliver value in a constantly changing environment. Differentiate the old marketplace and the new marketplace of the Information Technology. (6 MARKS)
 - d. A remote-first IT advisory firm, NovaBridge Consulting, delivers DevOps optimization and data privacy services to clients worldwide. When a critical zero-day vulnerability triggers widespread system outages across several customer environments, NovaBridge quickly mobilizes its distributed rapid-response squad. Using agile sprints, real-time collaboration channels, and automated deployment pipelines, the team rolls out patches, mitigations, and monitoring enhancements within the same day. By combining a virtual operating model with adaptive execution, NovaBridge restores stability fast, deepens client confidence, and earns new engagements through strong referrals. (9 MARKS)

Analyze the scenario above and explain the basic business strategies that the virtual company can use in their operation.
2.
 - a. Briefly explain the **THREE (3)** basic functions of a system. (6 MARKS)
 - b. Briefly discuss the term 'data' and 'information'. (4 MARKS)
 - c. The business use of IT refers to applying technology to strengthen day-to-day operations, support better decision-making, and deliver greater value to customers. By automating and improving processes, IT helps organizations use resources more effectively, boost efficiency, reduce costs, and gain a competitive edge. It also fuels innovation by enabling new products, services, and business models, while improving customer interactions and supporting data-driven strategies. (8 MARKS)

Differentiate at least **FOUR (4)** ways to implement the basic strategies in the business use in Information Technology.

- d. GlowFit, an online fitness subscription platform, introduced an AI-powered personalization engine that tailors workout plans and nutrition suggestions based on each user's goals, activity patterns, and in-app behavior. By leveraging IT and analytics, GlowFit made it easier for members to find relevant programs, improved engagement, and lifted upgrade conversions—raising average revenue per user by 18%. This smart use of technology enhanced convenience and results for customers while strengthening retention and driving overall growth. **(7 MARKS)**

Analyze the given scenario and discuss the best practices a company can apply to enhance customer values.

- 3. a. Briefly explain the importance of understanding the ethics of the Information System profession. **(6 MARKS)**

- b. Briefly describe at least **TWO (2)** common ethical issues in Information Systems. **(4 MARKS)**

- c. The digital divide is the gap between groups or areas that have reliable access to modern information and communication technologies (like the internet, smartphones, and computers) and those that don't. This imbalance can widen social and economic inequality by limiting people's access to education, employment, government services, and opportunities for growth. **(9 MARKS)**

In a rural district, many households have weak internet coverage and can't afford laptops. When a school shifts to online learning for assignments and exams, students in that area struggle to attend live classes, download materials, or submit work on time. Meanwhile, students in nearby urban neighborhoods with stable broadband and personal devices continue learning smoothly—leading to a growing gap in grades, skills, and future opportunities.

Analyse the paragraph above and describe the ethical dilemmas of digital divide.

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- d. Centralized data control means keeping an organization's data storage, management, and access in one primary system or location, usually governed by a single owner or team. This setup streamlines administration, ensures consistent access rules, and promotes uniform data standards across the organization. Tools like Google Drive and enterprise platforms from Salesforce and AWS support centralized control, improving accessibility, collaboration, and oversight. (6 MARKS)

Based on the given statement, compare the advantages and disadvantages of centralized data control.

4. a. Briefly describe **THREE (3)** types of system abuses. (6 MARKS)
- b. Briefly discuss the main approaches for minimizing vulnerability. (4 MARKS)
- c. Security controls in risk management are the safeguards an organization puts in place to recognize, reduce, and handle risks affecting its systems and data. They typically fall into three groups—preventive measures to stop issues before they occur, detective measures to spot threats or weaknesses, and corrective measures to respond and recover quickly. By applying these controls, organizations can protect critical assets, meet regulatory requirements, and sustain reliable operations. (6 MARKS)

Based on the given statements, analyze the business values for a company to invest in security.

- d. Fighting cybercrime requires strong safeguards to defend systems, networks, and information from malicious activity. Organizations rely on protections such as firewalls, intrusion detection tools, and encryption to block unauthorized access and reduce risk. Effective efforts also depend on cooperation among governments, businesses, and law enforcement to investigate and prosecute offenders. Training and awareness initiatives help people spot and handle threats like phishing and malware. Ongoing monitoring, timely patching, and well-practiced incident response plans improve an organization's ability to withstand constantly changing attack methods. (9 MARKS)

Analyze the paragraph above and discuss **FOUR (4)** methods for combatting cybercrime. Additionally, identify **ONE (1)** law or regulation that is used for cybercrime prevention.

-END OF MARKING SCHEME-